

INTEGRATION GUIDE

Twilio SMS & Voice Setup

Complete implementation, local development, and deployment walkthrough for PlowPath

This document provides the concrete steps to configure your Twilio account, set up local testing environments, and provision credentials to deploy the SMS notifications and interactive voice response (IVR) flows for the PlowPath operations platform.

 PRIORITY ACTION REQUIRED

Start A2P 10DLC brand registration immediately. Carrier reviews for Application-to-Person campaigns currently take **2 to 4 weeks**. Setting this up on day one prevents automated SMS notifications from being blocked in production.

1. Platform Architecture Invariants

The PlowPath backend is engineered to handle Twilio gracefully depending on your active environment:

- **Mock Mode (Default):** If Twilio keys are omitted from your environment variables, the system executes in a dry-run state. Outbound SMS texts are output directly to the backend logging stream, and voice calls operate under fallback modes. No API fees are charged.
- **Live Mode:** When Twilio keys are detected in your system environment, the live Twilio Node SDK is initialized and real SMS messages are routed to client devices.

2. Twilio Portal & Regulatory Setup

1. **Create & Fund Your Account:** Sign up at [twilio.com](https://www.twilio.com). Add a standard payment method and top up your account balance with a starting credit of \$20.
2. **Brand Registration (A2P 10DLC):** Navigate to **Console → Messaging → Regulatory Compliance → Brand Registration**. Submit your official business identification information (EIN/Tax ID, address, legal name, website). Brand approval typically takes 1–3 business days (costs a \$4 one-time carrier fee).
3. **Campaign Submission:** Under your approved Brand, create a new **Campaign**:
 - **Use Case:** `Mixed` (Notifications + IVR) or `Customer Care`.
 - **Sample Messages:** Add real transactional examples such as:
"PlowPath: Crew arriving at 123 Maple St in ~30 min. Reply STOP to opt out."
 - **Opt-in Flow Description:** Describe the user permission flow clearly:
"Customers opt in when signing the snow removal service contract. They consent to SMS notifications about service status."

Carrier campaign reviews take **1–3 weeks**.
4. **Purchase a Service Phone Number:** Go to **Console → Phone Numbers → Buy a Number**. Search for a local number in your business area (e.g., Buffalo `+1 716` area code). Ensure that it supports both **SMS** and **Voice**. This typically costs ~\$1.15 per month.
5. **Link Number to Campaign:** Associate your newly active phone number with your approved A2P 10DLC Messaging Campaign to enable secure messaging delivery.

3. Local Development Configuration

To run and test Twilio services locally on your machine, retrieve your account credentials from your Twilio Console dashboard and configure them in your local environment file.

A. Configure Local Env File

Add these entries in `backend/.env`:

```
# Twilio API Integration
TWILIO_ACCOUNT_SID=ACXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
TWILIO_AUTH_TOKEN=your_auth_token_here
TWILIO_PHONE_NUMBER=+1716XXXXXXX
```

B. Configure Local Tunnel for Webhooks

Twilio sends webhook events when inbound calls or text status updates occur. Because local servers are behind private networks, you must use a secure tunnel such as **ngrok** to expose your local port 3000 to the public web:

```
ngrok http 3000
```

Copy the generated public URL (e.g., `https://a1b2-cd34.ngrok-free.app`).

C. Update Twilio Webhook Endpoints

In the Twilio Console under **Phone Numbers → Active Numbers → [Your Number]**, scroll down to configurations and set the following endpoints (replacing the prefix with your active tunnel URL):

- **Voice / Call Comes In:** `https://your-tunnel.ngrok-free.app/api/v1/webhooks/twilio/voice` (POST)
- **Messaging / Message Comes In:** `https://your-tunnel.ngrok-free.app/api/v1/webhooks/twilio/sms` (POST)
- **Messaging / Status Callback:** `https://your-tunnel.ngrok-free.app/api/v1/webhooks/twilio/sms-status` (POST)

4. Production Deployment & Secrets Management

When launching the server environment into production on **Fly.io**, use secure secret provisioning rather than committing keys to your git codebase.

A. Set Fly Secrets

Run the following command within your local terminal from the `backend` directory:

```
fly secrets set TWILIO_ACCOUNT_SID="your_live_sid" \  
               TWILIO_AUTH_TOKEN="your_live_token" \  
               TWILIO_PHONE_NUMBER="+1716XXXXXXX"
```

B. Configure Production Webhooks

Update your Twilio active phone number settings in the Console to use your live production server API subdomains:

- **Incoming Voice URL:** `https://api.plowpath.app/api/v1/webhooks/twilio/voice`
- **Incoming SMS URL:** `https://api.plowpath.app/api/v1/webhooks/twilio/sms`
- **SMS Status Callback URL:** `https://api.plowpath.app/api/v1/webhooks/twilio/sms-status`

5. System Health Check Verification

Verify that your setup is alive and reachable by launching the local backend and hitting the status endpoint:

```
curl http://localhost:3000/health/twilio
```

A functional and successfully authenticated environment will respond with:

```
{  
  "status": "ok",  
  "twilio": "connected",  
  "apiStatus": 200  
}
```

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